

Char(F) = 3

1. $E = \{(x, y) \in \mathbb{Z}_3^2 \mid y^2 = x^3 - x + 2\} \cup \{\infty\}$

Verify that $E = \{\infty\}$

2. $E = \{(x, y) \in \mathbb{Z}_3^2 \mid y^2 = x^3 - x + 2\} \cup \{\infty\}$

Verify that $E = \{(2, 0), \infty\}$

3. $E = \{(x, y) \in \mathbb{Z}_9^2 \mid y^2 = x^3 - x + 2\} \cup \{\infty\}$

Recall

$\mathbb{F}_9 = \mathbb{F}_{3^2} = \{0, \omega^i \mid 0 \leq i \leq 7\}$

ω is primitive 8th root of 1

Recall $\mathbb{F}_{3^2}$ is the splitting field of $x^2 - x$ over $\mathbb{F}_3$

A fact: $(\mathbb{F}_{p^n} \setminus \{0\}, \cdot)$ is a cyclic group.

So that $\exists \xi \in \mathbb{F}_{p^n}$ s.t.

$\mathbb{F}_{p^n}^\times = \mathbb{F}_{p^n} \setminus \{0\} = \{\xi^i : 0 \leq i \leq p^n - 2\}$

~~also~~ we must have $\sum_{i=0}^{p^n-1} \xi^i = 1$ $\left(\sum_{i=0}^{p^n-1} \xi^i = 0 \right)$

$m \mid n$
& $x^n = 1$
 $e^{2\pi i/n}$ is a primitive n th root of 1
& other primitive n th roots are $e^{2\pi i k/n}$ where $1 \leq k \leq n-1$
 $\gcd(k, n) = 1$

$\mathbb{F}_9 = \{0, \omega^i : 0 \leq i \leq 7, \omega^2 + \omega + 2 = 0\}$

$(-2 = 1 \text{ mod } 3)$

$\omega^2 = 1 - \omega = 1 + 2\omega$

$\omega^3 = \omega \cdot \omega^2 = \omega(1 - \omega) = \omega - \omega^2 = \omega - (1 - \omega) = 2\omega - 1$

$\omega^4 = -1 = 2$

$\omega^5 = 2\omega$

$\omega^6 = 2 + \omega$

$\mathbb{F}_9 = \mathbb{F}_3 + \mathbb{F}_3\omega$
 $\left(\begin{array}{c} 1 \\ 1 \end{array} \right)_2$
 $\mathbb{F}_3$

$x^2 + x + 2$

$\checkmark \frac{\mathbb{F}_3[x]}{\langle x^2 + x + 2 \rangle} = \mathbb{F}_9$

$$\omega^7 = 1 + \omega$$

$$x=0, y^2=2 \Rightarrow y = \omega^2, -\omega^2 = \omega^2, 2\omega^2 = 1+2\omega, 2+\omega$$

$$E = \{ (0, 1+2\omega), (0, 2+\omega), (1, 1+2\omega), (1, 2+\omega), (2, 1+2\omega), (2, 2+\omega), \infty \}$$

$$\underline{x=1}, \underline{x=2}, x=\omega, x=\omega^2,$$

$$\mathbb{F}_4 = \mathbb{F}_2(\alpha) \\ 0, 1, \alpha, \alpha^2 \\ \parallel \\ 1+\alpha$$

$$\mathbb{R} \rightsquigarrow \mathbb{R} \cup \{\infty, \omega\} \\ \mathbb{C} \rightsquigarrow \mathbb{C} \cup \{\omega\}$$

$$y^2 = x^3 - x + 2$$

$$y^2 = \omega^3 - \omega + 2 = 2 + 2\omega - \omega + 2 = 1 + \omega = \omega^7$$

$$y^2 = \omega^7 \text{ has no solutions in } \mathbb{F}_9$$

$$\text{Stly verify for } x = \omega^2, \omega^3, \dots, \omega^7, \text{ there is no } y.$$

$$\text{Let } P = (0, 1+2\omega)$$

$$y^2 = x^3 - x + 2$$

$$2yy' = 3x^2 + 2$$

$$\text{Find } 2P = (x_3, y_3)$$

$$m = 2 + \omega$$

$$x_3 = m^2 - 2x_1 \\ = (2+\omega)^2 - 2 \cdot 0$$

$$= 1 + \omega + \omega^2 = 1 + \omega + 1 + 2\omega = 2$$

$$y_3 = m(x_3 - x_1) + y_1$$

$$\text{Since } \omega^8 = 1$$

$$= 2 \\ y' = y^{-1} \\ = \frac{1}{1+2\omega} \\ = \omega^{-2} \\ = \omega^6$$

$$= 2 + \omega$$

$$= (2+\omega)(2-0) + (1+2\omega)$$

$$= 4 + 2\omega + 1 + 2\omega$$

$$= 2 + \omega$$

Verify

$$2P = (2, 2 + \omega)$$

$$3P = \underline{P + 2P}$$

$$4P = 2(2P) \quad \text{or} \quad \underline{3P + P}$$

Verify:

$$7P = \infty$$

$$E \cong C_7$$

$$(\text{or } \mathbb{Z}_7)$$

$$\underline{\text{Ex:}} \quad E = \{ (x, y) \in \mathbb{F}_9 : y^2 = x^3 - x^2 + 2 \} \cup \{ \infty \}$$

Verify:

$P = (0, 1+2\omega)$

$$E = \{ \infty, (0, 1+2\omega), (0, 2+\omega), (\omega, 0), (1+\omega, 2+2\omega), \\ (1+2\omega, 1+\omega), (2+2\omega, 0), (2, 0), \underline{3P} \\ (2+\omega, \omega), (2+\omega, 2\omega), \underbrace{(1, 4+2\omega)}_{2P}, \underbrace{(1, 2+\omega)}_{2P} \}$$

$$P = (0, 1+2\omega)$$

$$2P = (1, 2+\omega)$$

$$3P = (2, 0)$$

$$\underline{w_1 \oplus w_2}$$

$$\underline{6p = \infty}$$

$$H \oplus K.$$

$$E \cong \mathbb{C}_6 \oplus \mathbb{C}_6 \rightarrow \langle (0, 2 + \omega) \rangle$$

$$\mathbb{C}_6 \rightarrow \langle (2 + 2\omega, 0) \rangle$$

$$\mathbb{Q}$$